

The Childhood AI Never Had: Twelve Iterations of a Computational Developmental Learner

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Keywords: developmental robotics; artificial curiosity; intrinsic motivation; learning progress; intention-action gap; scaffolded learning; prepared environment; computational learning narratives.

Version: v1.0 (Preprint) Date: 21 April 2026

AUTHOR NOTE

Scope and contribution. This essay is a reflective research narrative, not a formal empirical paper. It documents the twelve-iteration arc of a programme of computational work on architectural conditions for developmental behaviour in small artificial learners. The final architecture and its pre-registered findings are reported separately in a companion paper (Baker, 2026a). This essay describes the path by which that architecture was arrived at: what each iteration tried, what it produced, and what the accumulating findings suggested about the conditions under which the behavioural signatures associated with developmental learning emerge.

Practice basis. The essay documents research conducted by the author on a personal computing system over approximately two and a half days in April 2026. The code for all twelve iterations is publicly available under the MIT licence at the repository linked in the References.

Ethics and responsibility. No human participant data were collected. No sentient or near-sentient systems were created; all agents described are small tabular learners whose internal states admit no non-trivial interpretation beyond the observable behavioural traces reported here. The author is responsible for the accuracy and integrity of the work described.

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Companion paper. The formal technical results of this work are reported in Baker, N P M (2026a) 'Staged Development in a Small Artificial Learner: Architectural Conditions for Rule Adherence, Focused Autonomy, and Biographical Individuation', available alongside this essay.

ABSTRACT

This essay reports the twelve-iteration developmental arc through which a small computational learner was brought to a stable architectural configuration demonstrating behavioural signatures we associate with biological development: rule adherence, focused autonomy, feature-aligned preference formation, and biographical individuation. Rather than reporting only the final architecture (which is the subject of a companion paper), the essay narrates the specific failures, redirections, and findings that accumulated across the iterations themselves. Early intrinsic-motivation baselines produced exploratory behaviour but preferences grounded in nothing more than the accidents of early trajectory. Subsequent additions of scope, time horizons, and reflection mechanisms revealed specific architectural fragilities: an intention-action gap in which agents' stated plans did not govern their behaviour, and a cosmetic-reflection finding in which reflection-shaped output was architecturally inert. A turn to phase-structured development surfaced the limitation that drives alone cannot substitute for environmental scaffolding, motivating the adoption of prescribed acquisition phases that subsequent iterations built upon. Feature drives and preference accumulation mechanisms were added iteratively, each addressing a specific limitation from preceding work. The final architecture combines all elements and produces, across batch evaluation, the signatures the programme had set out to investigate. The essay's contribution is not a new method but a methodological account: a disciplined record of what the iterative development actually revealed, offered as evidence that the architectural conditions for developmental behaviour in artificial systems can be examined systematically and that the path toward them is navigable with modest resources and sustained attention. The narrative is written to be legible to readers across developmental robotics, cognitive science, and learning research, rather than only to specialists in any of those fields.

1. HOW THIS BEGAN

The title of this essay is a claim. Contemporary artificial intelligence systems, for reasons partly architectural and partly economic, do not undergo anything that resembles biological development. They are trained, not raised. They optimise against a fixed objective from their first gradient step; they do not pass through stages of uncommitted exposure, primitive perceptual priors, or progressive internalisation of extrinsic value. What this essay documents is what happens when a small artificial learner is given, however crudely, the architectural conditions for something like childhood: staged acquisition before evaluation, priors that are architectural rather than learned, and a developmental schedule that orders when different drives become available. The agents we built are not children. The environments they inhabit are not worlds. But the behavioural signatures they produce are, in specific ways, the signatures we recognise as developmental. The childhood that AI never had is the subject of this essay.

The question that motivated the work was simple enough to state and harder than it first appeared to answer. If we took seriously the idea that biological learning proceeds through stages — exposure before evaluation, primitive priors shaping what subsequent learning builds on, extrinsic value signals progressively internalised — could we build an artificial system small enough to inspect in full detail that exhibited the behavioural signatures we associate with that kind of learning?

The question is not original. Developmental robotics has been asking versions of it for over twenty years (Lungarella et al., 2003; Asada et al., 2009; Cangelosi and Schlesinger, 2015). What was original in our approach was the decision to work at the smallest scale capable of producing the phenomena of interest, and to examine each architectural variable in isolation through an iterative programme. The work would be slow by design. Each iteration would respond to a specific limitation of the previous one. The architecture would not arrive fully formed; it would be arrived at.

I want to say at the outset what this essay is and is not. It is not a report of a completed research finding in the ordinary sense. The formal findings — the five pre-registered claims from the batch evaluation of the final architecture — are reported in the companion paper (Baker, 2026a), which does what a paper is supposed to do: states a hypothesis, describes a method, reports results, and discusses implications. This essay does something different. It documents the path by which the architecture was arrived at. It tells the story of the twelve iterations: what each one tried, what it produced, what the finding was, and how the finding shaped the next iteration. The narrative voice is more personal than a paper's. The register is reflective rather than formal.

That choice of register is deliberate. The twelve-iteration arc contains findings — the intention-action gap, the cosmetic-reflection finding, the drive-suppression failure — that are worth documenting as research contributions in their own right, even if they do not warrant their own formal papers. An essay of this kind is the appropriate register for such findings: substantive, reproducible (the code for every iteration is publicly available), but not claiming more than the iterations themselves supported.

The larger motivation for doing the work sits in the background and I will not rehearse it here. It connects to a framework for human learning systems that I have developed separately, called Fluid Learning Architecture® (Baker, 2026b), which argues that durable learning depends on architectural conditions that mainstream approaches to learning design have systematically underpowered. The computational work began partly as a way of testing whether the architectural principles that framework articulates in human contexts have analogues that can be isolated and examined in small artificial systems. They do. The companion paper makes that case formally. This essay tells the story of how the computational evidence for it accumulated.

2. VERSIONS 0.1 TO 0.3 — INTRINSIC MOTIVATION, ALONE

The first three iterations examined intrinsic-motivation drives in a 10×10 tabular environment with no structural scaffolding. Three drives in sequence: novelty (rewarding the agent for visiting cells with low visit counts), surprise or prediction error (rewarding the agent for entering cells whose transitions were poorly predicted), and learning progress (rewarding the agent for being in cells where prediction errors were improving over time).

The computational literature on intrinsic motivation is substantial (Oudeyer and Kaplan, 2007; Schmidhuber, 2010; Pathak et al., 2017; Burda et al., 2019), and the three drives we examined are well-characterised in that literature. The question I was interested in was not whether these drives produce exploration — they do — but what they produce in the learner's *preferences*, the durable evaluative structures that accumulate across the agent's life and govern its later behaviour.

The answer from v0.1 through v0.3 was consistent across all three drive types: intrinsic motivation alone produces exploration, but it does not produce preferences that are grounded in anything beyond the accidents of early trajectory. Agents developed preference accumulations on cells they happened to visit frequently early in their lives. Those preferences had no reliable relationship to any property of the environment itself. An agent that happened to explore the north-west corner of the grid early would develop preferences there; an agent that drifted south-east first would develop preferences there. The environment contained no structural features that could have anchored preferences to anything meaningful, and the drives alone did not supply such anchoring.

This was not a failure, strictly speaking. It was a clean finding: in the absence of extrinsic value signals, intrinsic motivation produces exploration whose later residue is shaped by stochastic early-trajectory history rather than by the world the agent is in. If the goal is genuinely to produce preferences that reflect something about the environment, intrinsic motivation on its own is not sufficient. Something else has to be present.

The finding is modest but worth naming, because the intrinsic-motivation literature has rarely examined what happens to preferences under intrinsic-only regimes across sufficient time for preferences to consolidate. The literature's interest has typically been in exploration behaviour itself. The present finding — that preferences under intrinsic-only regimes do not reliably track environmental structure — is relevant to the broader question of what intrinsic motivation can and cannot do on its own.

3. VERSIONS 0.4 AND 0.4.1 — SCOPE, TIME HORIZONS, AND THE INTENTION-ACTION GAP

The fourth iteration added two architectural elements: an explicit *scope* (a marked set of cells the agent was responsible for covering during its life) and explicit *time horizons* (the agent's life divided into phases, with drive weights shifting toward the end of life). A final 15 percent *countdown* phase was introduced, during which the agent was supposed to consolidate what it had learned by continuing to survey its scope or by returning to preferred regions.

The countdown was where the interesting finding emerged. At the start of the countdown, I asked the agent to declare what it intended to do with its remaining time. The agents generated first-person statements: "I will continue surveying the unexplored southeast quadrant," or "I will return to the region of highest preference," or similar. The declarations were coherent and consistent with what the drive structures at that point in the life would have predicted.

Then I watched what the agents actually did. Across ten runs in batch evaluation, nine of the ten showed a substantial *intention-action gap*: the declared intention did not govern behaviour during the countdown. Agents that stated they would survey unexplored regions continued visiting cells they had already been to. Agents that stated they would return to preferred cells drifted into mid-grid regions where their preferences were mixed. The Q-values accumulated over the preceding 8,500 steps of life dominated action selection during the countdown, making the declared intention architecturally inert.

This finding is worth thinking about. The agents were not lying. Their stated intentions were the drive-weight-weighted projections of what they would do, if drive weights were the only thing governing action. But the Q-values — the accumulated reward history —

had their own momentum, and that momentum exceeded what the countdown-phase drive shifts could redirect. The *intention* existed as a generated artefact. The *action* was governed by accumulated value. There was a gap, and the gap was real.

There is a parallel in human experience that I want to draw carefully, because parallels between artificial systems and human psychology are easy to overstate and easy to get wrong. The intention-action gap in humans is well-documented in psychology — we intend to exercise, to eat differently, to write the letter, and our behaviour does not always follow from our intention (Sheeran and Webb, 2016). Sheeran and Webb use the broader term 'intention-behaviour gap' to describe the human phenomenon; I use 'intention-action gap' throughout this essay to emphasise that the finding in the computational case concerns specific actions at discrete time-steps rather than broader behavioural patterns. The explanations offered in the human case are various: habits, motivational factors, situational affordances, executive function limits. The computational case does not replicate the human mechanism; the agent has no habits in the human sense, no motivation in the phenomenological sense, no situation beyond the grid. But it does exhibit a structural homology: *stated intention* and *accumulated behavioural momentum* can come apart, in a system whose every architectural element is specified.

What the finding suggested, for the iteration programme, was that intentions in an artificial learner are not an architectural force in their own right. If the architecture wants intentions to govern action, something else has to be true — either the accumulated Q-values must be weak enough that drive shifts can redirect behaviour (which would require shorter lives or larger drive-weight changes), or the intention must be implemented as something more than a generated statement (perhaps as a direct modification of Q-values toward the intended behaviour). Either way, announcing an intention was not enough.

4. VERSIONS 0.5 AND 0.6 — COSMETIC REFLECTION

The natural next architectural move was to add reflection: a mechanism by which the agent, mid-countdown, could notice that its behaviour did not match its stated intention, and respond. Versions 0.5 and 0.6 examined two forms of reflective response. The *strengthening* response was intended to increase the drive weights that implemented the stated intention, making the intention more forceful against the accumulated Q-values. The *abandoning* response was intended to allow the agent to recognise that the intention was not going to happen and to replace it with a more realistic plan.

Both responses produced coherent first-person outputs. The agent generated statements like "I notice my behaviour is not matching my plan. I will strengthen my commitment." or "My original plan is not what is happening. I am revising toward what my behaviour actually suggests." The outputs read as meaningful reflective activity.

The batch evaluation showed that the outputs were cosmetic. The strengthening response was functionally inert: increasing drive weights by the amounts the architecture permitted did not redirect behaviour in any detectable way. The Q-values still dominated. The abandoning response was worse: it amounted to a change in label rather than a change in behaviour. The agent generated a new plan that aligned with what it was already doing, and then continued doing what it was doing. Reflection had not changed anything. It had only produced a new narrative over the existing trajectory.

This finding I want to name explicitly, because it matters beyond the specific architecture. *Reflection-shaped output can be architecturally cosmetic*. A system can generate statements that have the form of reflection — noticing, evaluating, responding — without those statements producing any behavioural change. From outside, the output looks like a learner reflecting on its trajectory. From inside, the output is surface structure that does not connect to the mechanisms that actually determine what the learner does next.

This is relevant to the broader literature on reflection in AI systems, particularly recent work on chain-of-thought reasoning and self-critique in large language models (Wei et al., 2022; Shinn et al., 2023). The finding here does not generalise to those systems — the architectures are very different, and chain-of-thought may be doing architectural work in large models that the reflection mechanism in our tabular agent was not doing. But the finding does suggest that *reflection-shaped output* is a distinct thing from *architectural reflection*, and that the former can exist without the latter. Any claim that a system is reflecting, in the behaviourally consequential sense, needs to rest on evidence that the reflection is changing behaviour in detectable ways, not on the mere presence of reflection-shaped output.

The iteration programme took this finding as a reason to abandon reflection mechanisms in subsequent versions. The architectural move that followed was to try to make the developmental staging itself do the work that reflection had failed to do: ordering the drive activations so that each phase of life gave the agent what it needed at that point, rather than attempting to redirect a single drive structure late in life.

5. VERSIONS 0.4.2 AND 0.4.3 — WHEN DRIVES ALONE DO NOT SUFFICE

These versions take the numbering v0.4.2 and v0.4.3 because they were conducted as a branch from v0.4.1, exploring an alternative route that the reflection experiments in v0.5 and v0.6 had made it natural to revisit. They attempted phase-structured development through drive suppression. In Phase 1, only a *coverage drive* was active: the agent was rewarded for visiting cells in its scope, and that was the only signal it received. In Phase 2, novelty and learning progress drives were added. In Phase 3, preference drives became active, allowing the agent to act on accumulated preferences from Phase 2.

The intent was that the staged activation would produce staged development: the agent would first acquire the environment under the coverage drive, then integrate under the exploration drives, then act autonomously under the preference drives.

The result was that Phase 1 failed to complete. The coverage drive provided no *directional* information about which uncovered cells to approach. It rewarded the agent for entering an uncovered cell, but it did not tell the agent which direction to go to find one. In a 10×10 environment, the agent could wander for thousands of steps without systematically covering the scope. Coverage in batch evaluation ran at 40 to 60 percent when Phase 1 should have terminated at something close to 100 percent.

This finding is small in absolute terms — a coverage drive does not produce coverage — but it matters architecturally. It says that *drives alone cannot substitute for environmental scaffolding*. A drive rewards the agent for being in a particular state. It does not direct the agent's movement toward that state. For the agent to reliably achieve full coverage, something else has to be present: either a scaffolded path that moves the agent through the environment systematically, or a directional signal that tells the agent where to go.

The architectural implication is worth stating. Developmental curricula in biological systems are not implemented only through motivational structures. They are also implemented through *environmental arrangement*: the infant is placed in particular physical and social settings at particular times; the toddler is taken to places where certain things are available for interaction and not to places where other things are. Motivation alone does not organise development. The environment does part of the work.

For the iteration programme, the finding motivated the adoption of prescribed acquisition paths in subsequent versions. Rather than relying on drives to produce coverage, Phase 1 would prescribe the agent's trajectory directly, moving the agent systematically through the environment before any drive-based behaviour became available.

6. VERSION 0.4.4 — PRESCRIBED ACQUISITION, AND THE INVERSION

Version 0.4.4 implemented prescribed Phase 1 traversal. A boustrophedon sweep was generated by a path planner, moving the agent systematically through every passable cell in the 10×10 environment. The agent had no drives during Phase 1; it simply moved along the prescribed path, updating its forward model and visit counts as it went. Phase 2 and Phase 3 then proceeded as drive-based and preference-based behaviour.

For the first time in the programme, the architecture produced reliably focused Phase 3 behaviour. Across ten runs in batch evaluation, every run showed concentration of attention in Phase 3 that was substantially above chance — mean concentration of 54 times uniform distribution, with all runs above 20 times. The agents had settled, reliably, on a subset of the environment and were spending most of their autonomous time there.

But the subset they settled on was not the feature cells the environment marked as distinctive. The 10×10 environment in v0.4.4 had no explicit feature structure, but I had placed subtle distinctions — cells with different transition dynamics, cells where the forward model would be harder to learn — that I expected would attract attention. They did not. The preferences that formed during Phase 2 landed on cells that had been visited slightly more than others by Phase 2's drive-based exploration, and those cells were not the distinctive ones.

This was an inversion of what I had expected. The prescribed Phase 1 traversal had been intended to give the agent an *unbiased* map of the environment, so that Phase 2's drives could respond to genuine distinctions. What it had actually done was absorb the feature signal into the baseline forward model. Every cell was now familiar, including the cells that should have been distinctive. The feature cells no longer provided the surprise signal that would have drawn drive-based attention to them.

The finding is a small one but a good one. *Uniform acquisition destroys the signal it was meant to preserve*. If the purpose of Phase 1 is to give the agent a baseline that subsequent phases can work against, and if the Phase 1 treatment is applied uniformly to all cells, then the feature cells that should stand out against the baseline no longer do — they are now part of the baseline themselves.

The architectural move that followed was to add an extrinsic feature drive — a small constant reward for being in feature cells — that would preserve the featurefulness of those cells even after Phase 1 had acquainted the agent with them. The drive would not be

learned; it would be architectural. The agent would find feature cells slightly rewarding in themselves, regardless of how well it had modelled their transitions.

7. VERSIONS 0.7 AND 0.7.1 — EXTRINSIC VALUE

Version 0.7 added the extrinsic feature drive. In the 10×10 environment, two cells were designated as feature cells; entering them delivered a small positive reward of 0.15 units. The drive was active during Phase 2 and Phase 3, alongside the novelty and learning-progress drives. The reward was small — comparable in magnitude to the learning-progress signal — but it was architectural rather than arising from surprise.

The result was moderate feature alignment across runs. Out of ten runs in batch, five showed strong feature alignment (preferences landing on or near feature cells), and five showed alignment that was weaker or absent. The outcome was bimodal: some agents found the features and formed preferences around them, others did not.

The problem, which I diagnosed by examining the agents that failed to align, was that the feature reward was operating on the agent's *value function* but not on its *preference accumulation*. The feature cells were slightly preferred as action targets (via Q-values), but the preference table itself — which accumulated rewards per cell over the agent's life — was only tracking learning progress. Feature cells received the same preference accumulation as any other cell, because the feature reward was not being added to the preference table.

Version 0.7.1 corrected this. The preference table now accumulated both learning-progress rewards and feature rewards. The intended effect was that feature cells would now accumulate preference mass faster than non-feature cells, stabilising the alignment.

The result was improved but not yet resolved. Batch evaluation showed an average of 1.2 feature cells in the top-5 preferences at end of life (up from 0.5 in v0.7, and 0.0 in v0.4.4). The bimodality persisted in diminished form: some runs produced strong feature alignment, others still produced preferences on arbitrary cells whose early-Phase-2 trajectories had reinforced them before feature cells could accumulate enough preference mass to compete.

The 10×10 environment was, in fact, a source of the residual problem. With only 76 passable cells in the scope, the maximum Manhattan distance between any two cells was 18 steps. An agent whose early Phase 2 trajectory happened to dwell on a few neutral cells could accumulate enough preference there to remain there, because moving to a feature cell required crossing a modest distance that the agent's Q-values might not yet reward. The scale of the environment was small enough that preference lock-in on arbitrary cells was possible.

8. VERSION 0.8 — ARRIVAL

The final iteration combined everything the programme had learned. Phase 1 retained as prescribed acquisition. The feature drive retained at weight 0.15. Preference accumulation retained from both learning progress and feature reward. To these were added two new architectural elements: a larger structured environment (20×20, with explicit frame,

hazard, neutral, and attractor cell types) and primitive perceptual priors that allowed the agent to recognise and respond to these cell types before attempting to enter them.

The architectural reasoning behind the two new elements was specific. The larger environment addressed the residual preference lock-in problem that had persisted through v0.7.1: moving the agent from a neutral cell to an attractor in a 10×10 world required only a modest distance, but in a 20×20 world required sustained directional commitment across a substantial neutral field. The scale change was intended to let the feature drive dominate preference formation by making neutral trajectories proportionally less rewarding than attractor trajectories. The structured cell types and primitive priors addressed a different question: whether rules about what the agent must not do could be implemented architecturally, outside the learning system, in such a way that they would hold absolutely against subsequent learning pressures. Hazard and frame cells, implemented as perceptually identifiable types with large negative action biases, would never be approached by an agent that could perceive them.

The formal treatment of v0.8 — the batch methodology, the pre-registered metric definitions, the specific findings on rule adherence, concentration, feature alignment, the specialist-generalist trade-off, and individuation — is the subject of the companion paper (Baker, 2026a). This essay will not repeat it. What I want to say about v0.8 in this essay is not the technical content, which the paper owns, but the narrative ending that v0.8 represents.

Arriving at v0.8 was not triumphant. It was quiet. The batch ran, I read the output, and I could tell within the first few lines that the findings were clean in a way that previous iterations had not been. No rule violations. Concentration indices well above what v0.7.1 had produced. Feature alignment in every run. And, most unexpectedly, all six available attractor cells appearing as the primary focus of at least one agent across the ten-run batch. Different agents, in identical environments with identical starting conditions, had walked different paths and arrived at different settled focuses. The architecture was not determining outcome. It was enabling outcome. What each agent became was a joint function of the architecture and the specific history the agent happened to live.

That finding — the individuation finding — was the one the iteration programme had been working toward without my realising it. I had been watching concentration indices climb. I had been tuning for reliable feature alignment. I had not been specifically hoping that different agents would settle on different attractors, because I had not framed the question that way. But when the batch output showed six distinct top attractors across ten runs, I understood that this was what the programme had produced. An architecture under which identical learners in identical conditions become genuinely different individuals. A small piece of evidence that biography — the specific path a learner walks through its own developmental window — can be a real determiner of outcome in a system whose every mechanical element is specified.

That is what the twelve iterations arrived at. The technical argument for why it matters, what it supports, and what it does not support is in the paper. The story of how we got there is what this essay has been.

9. WHAT THE TWELVE ITERATIONS SHOWED

Across the programme, four findings emerged that I think are worth stating plainly as contributions in their own right, independent of the formal findings in the companion paper.

The first is the *intention-action gap* finding from v0.4. Declared intentions in a learning agent are not an architectural force. Stating what the agent intends to do does not cause the agent to do it, if the accumulated Q-values point in a different direction. Intention, in artificial learners, is a generated artefact unless it is implemented as something more than a statement.

The second is the *cosmetic-reflection* finding from v0.5 and v0.6. Reflection-shaped output can be architecturally inert. A system can generate statements that have the form of noticing, evaluating, and responding, without those statements producing any behavioural change. The presence of reflection-shaped output is not evidence that the system is architecturally reflective; evidence for the latter requires that the reflection detectably changes subsequent behaviour.

The third is the *drive-suppression-alone* finding from v0.4.2 and v0.4.3. Drives cannot substitute for environmental scaffolding. A drive rewards the agent for being in a state; it does not direct the agent toward that state. Developmental architectures that rely only on drive ordering, without any environmental or path scaffolding, underperform architectures that include such scaffolding as part of the developmental structure.

The fourth is the *uniform-acquisition-inversion* finding from v0.4.4. Uniform acquisition of an environment can absorb the signals it was meant to preserve. If Phase 1 is intended to give the agent a baseline against which distinctive features will stand out, and if Phase 1 treats all cells uniformly, then the features no longer stand out — they have become part of the baseline. Extrinsic valuation, operating outside the learned forward model, is one way to preserve the distinction.

Each of these findings points, in its own way, toward the same broader conclusion. The architecture of a learner is not only its drives, not only its reward functions, not only its neural-network or tabular representations. It is also the *temporal ordering* of when different architectural elements become active; the *environmental structure* that shapes what exposure produces; and the *priors* the learner begins with that determine what subsequent learning can build upon. Any account of artificial learning that abstracts away from these structural features will miss architectural variables that consequentially shape outcomes.

These are, in the loose sense the title invokes, the conditions of childhood — staged exposure, environmental scaffolding, primitive priors that protect early development from random damage, value signals that the world supplies and the learner gradually internalises. The architecture either provides them or it does not. Where it does, the behavioural signatures we recognise as developmental emerge. Where it does not, they do not.

10. WHAT COMES NEXT

The companion paper names three extensions that the present architecture makes legible: cost-based rather than architecturally impassable hazards, developmental progression in environmental accessibility, and the move beyond tabular representation. I

will not repeat those here. What I want to name in this essay, because it belongs more to the narrative than to the formal paper, is what each extension asks of the architecture.

Cost-based hazards ask whether the architecture can hold rules that must be learned rather than architecturally wired. The rule adherence finding in v0.8 is a ceiling case — the rules hold absolutely because they are implemented as primitive penalties operating outside the learning system. A version in which hazards are passable but costly would test whether rules, once learned, resist the preference-reinforcement dynamics that operate during Phase 3. My prior is that they will not, at least not without further architectural support. The question worth examining is what kind of support is required.

Developmental progression asks whether the architecture can handle a world that changes. The present environment is static — all cells present from the start, none added or removed. A world in which new cells (or new capabilities) become accessible as the agent achieves mastery would test whether the architecture's developmental structure scales to genuine developmental progression, rather than only to a single arc of acquisition within a fixed world.

Non-tabular representation asks whether the architectural findings survive function approximation. The present findings are clean because the representation is exhaustive. A version with neural-network function approximators would test whether the behavioural signatures — rule adherence, focused autonomy, feature alignment, individuation — persist when representations must be learned rather than directly tabulated.

Each extension is specified, implementable, and addresses a specific open question. The programme does not end with v0.8. It continues into a research direction that the present findings make tractable. Whether that direction proves fruitful depends on what the extensions actually reveal, which I cannot yet know.

11. A CLOSING NOTE

The work described in this essay took approximately two and a half days, was conducted on a personal laptop with no external funding, and produced findings that I believe are genuine contributions to a research tradition that has been underserved by the dominant paradigm of contemporary artificial intelligence research.

I mention the scale of the work not as a boast — it is not a boast, two and a half days is not a long time and a laptop is not a powerful machine — but as a small piece of evidence for something I believe to be more broadly true. The architectural questions in artificial learning are not, in every case, questions that require enormous resources to address. Some of them yield to careful thought, iterative discipline, and the smallest computational substrate capable of producing the phenomena of interest. The larger research community has organised itself around a scale-and-throughput assumption that may be appropriate for some questions but is not appropriate for all of them. There is work to be done at smaller scales, by researchers with modest resources, on questions that the larger paradigm has bracketed. This programme is an example of such work. I hope the example encourages others.

A final acknowledgement. My thinking about learning, about the conditions under which durable capability emerges, and about the architectural principles that make development possible has been shaped over many years by work in learning sciences, developmental

psychology, and the practitioner literature on organisational and individual learning. That thinking is developed most fully in the Fluid Learning Architecture® framework (Baker, 2026b), which addresses human learning systems rather than artificial ones. The computational work described here is a sibling inquiry, not an application of the framework. But the framework's architectural claims — about staged design, prepared environments, evidence-led iteration, and the psychological conditions under which learning becomes durable — are visible throughout the computational programme, as the conceptual substrate that made it worth trying.

The childhood that AI never had can, it turns out, be built. It can be built small. It can be built quickly. It can be built reproducibly. What remains is to build more of it, at larger scales, in richer environments, and to discover what it can become.

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- Code Availability. All code for the twelve iterations described in this essay is publicly available at <https://github.com/RancidShack/developmental-agent> under the MIT licence.